

Better Gate Model — Alt Training Report

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Dataset /home/malnitro5/Documents/Projects/tyndall/Ikmal/Gate Model/Better Gate Model/leaf_nonleaf_data

Overall accuracy (clean test)

96.47%

3334 / 3456

Camera-stress accuracy

95.92%

3315 / 3456

Best EMA validation

96.47%

Target

92%

PASS

Per-class accuracy (recall)

leaf

clean

94.5%

stress

96.4%

P=0.984

F1=0.964

n=1728

non-leaf

clean

98.4%

stress

95.5%

P=0.947

F1=0.965

n=1728

FN_leaf (true leaf → non-leaf)

95

FP_leaf (true non-leaf → leaf)

27

Student TinyLeafGate

27,538 params

96×96

Teacher MobileNetV3-Small

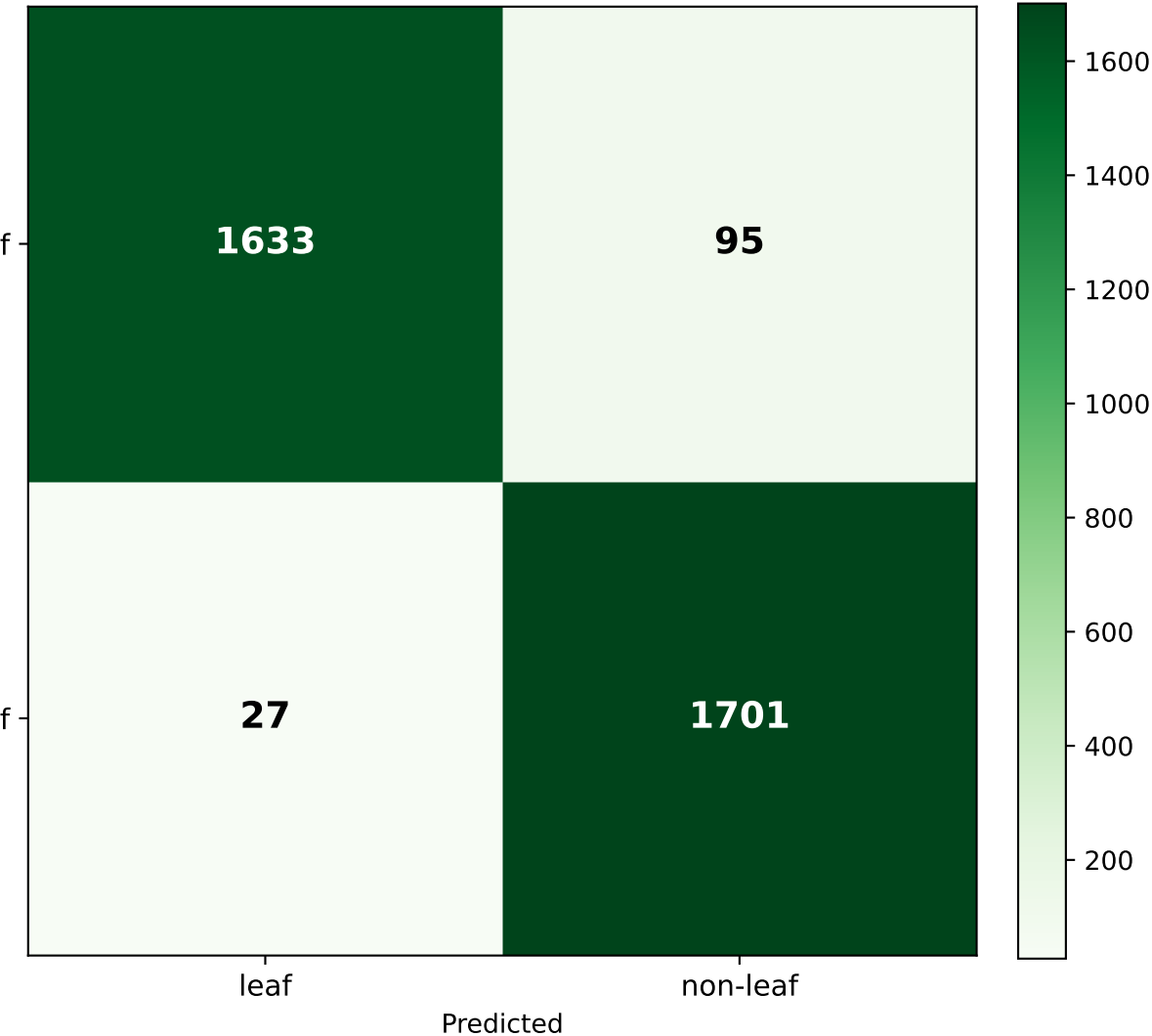
KD α=0.7

T=4.0

MixUp=0.2

EMA=0.999

Confusion matrix — clean PyTorch FP32 test



Dataset splits (balanced leaf / non-leaf)

Split	leaf	non-leaf	total
train	8,067	8,067	16,134
val	1,729	1,729	3,458
test	1,728	1,728	3,456

Pools were collected from PlantVillage first (leaf vs non-leaf),
balanced to the same count, then split into train/val/test.

Confusion matrix rows are true labels; columns are predictions.
Diagonal cells are correct counts.